Anas Ahmed

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Aspiring Machine Learning Engineer with a strong passion for technology, continuous learning, and applying AI to real-world problems. Proven track record of building deep learning models, engineering advanced ensemble pipelines, and deploying interactive applications.

EDUCATION

Alexandria University *Computer Science, specializing in Artificial Intelligence*
Timeline: October 2023 – Expected May 2027

EXPERIENCE

Internship at SIDPEC (Sidi Kerir Petrochemicals Company)

- Collaborated with technical teams to analyze and optimize data processing workflows used in large-scale industrial environments.
- Evaluated industrial computing systems and software applications to identify operational efficiency improvements.
- Analyzed the fundamental architectures of automation systems, control processes, and enterprise IT infrastructure.

Internship at ETHYDCO (Ethylene & Derivatives Company)

- Analyzed enterprise-level database architectures to understand how large-scale systems are designed, managed, and secured.
- Evaluated data protection strategies, risk-mitigation frameworks, and cybersecurity practices implemented in industrial environments.
- Investigated networking infrastructure, including enterprise switches and routing protocols, to ensure reliable plant-wide connectivity.

TECHNICAL PROJECTS

AI-Powered Children's Learning Platform – Speech & Computer Vision

- Developed a multimodal learning platform combining Whisper Small, Computer Vision, and adaptive learning for English pronunciation and handwriting.
- Built phoneme-level pronunciation and handwriting tracing pipelines using MFA, GOP, TensorFlow, and OpenCV.
- Implemented personalized learning paths and LLM-powered feedback using the Groq API.

Chest X-Ray Pneumonia Detection – Deep Learning (MobileNetV2)

- Trained and fine-tuned a MobileNetV2 model to classify chest X-ray images as Pneumonia or Normal.
- Applied data augmentation, batch normalization, and dropout techniques to improve model generalization.
- Deployed an interactive Streamlit web application for real-time predictions with confidence scores.

Stellar Classification – Advanced Heterogeneous Ensemble Stacking

- Built a 2-layer stacking pipeline using LightGBM, XGBoost, CatBoost, and a Ridge meta-model.
- Engineered custom astronomical features including color deltas and redshift scaling.
- Optimized final decision boundaries using Nelder-Mead to maximize Balanced Accuracy.

COURSES & CERTIFICATES

- **DeepLearning.AI – Machine Learning Specialization** Built a strong foundation in supervised and unsupervised learning; implemented regression models, neural networks, decision trees, and ensemble methods; applied model evaluation and bias–variance analysis.
- **Stanford CS230 – Deep Learning** Studied deep neural network architectures and the mathematics of backpropagation; implemented optimization techniques including Adam, RMSProp, and learning rate scheduling; applied regularization to mitigate overfitting.
- **Stanford CS231N – Deep Learning for Computer Vision** Focused on convolutional neural networks (CNNs) for image classification tasks; studied transfer learning, feature extraction, and modern CNN architectures on image-based datasets.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, SQL (MySQL, SQLite3)
Frameworks & Libraries: TensorFlow, Keras, Scikit-Learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy, Matplotlib, Flask, Streamlit
Tools & Technologies: Git, GitHub, VS Code, Linux, Cloud Computing
Concepts: Data Structures & Algorithms, Deep Neural Networks, CNNs, Sequence Models (GRU/LSTM), Feature Engineering, Memory Management